Nylon is very strong and can be used for rock climbing!

Nylon is a plastic polymer made of chemical units called polyamides. Polyamides are made with monomers – hexamethylenediamine and adipic acid. Solid chips of these polyamides are melted and forced through a heated spinneret which has very, very tiny holes.



Parachute



mountaineer

ACTIVITY 2

How Strong is Nylon?

Take an iron stand with a clamp. Take samples of cotton, wool, nylon and silk threads of about 50cm in length.

First tie cotton thread to the stand so that it hangs freely from it. At the free end, attach a CD as plate so that weights can be placed on it. Add weights starting from 10 grams one by one, untilthe thread breaks. Note down the total weight required to break the cotton thread. Repeat the same activity with the wool, silk and nylon threads.

NOTE: All the varieties of threads should be of same thickness.

Arrange the threads in the order of increasing strength.

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What do you infer from the above activity?

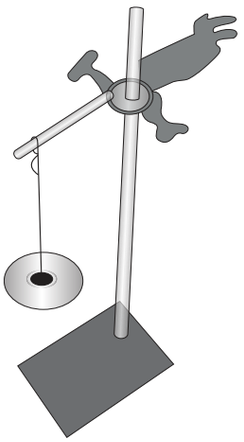
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Which type of fibre is the strongest?

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Which type of fibre is the weakest?

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Iron clamp with a hold weights

Polyester and Acrylic: Synthetic Fibres

Polyester is another synthetic fibre. It can be drawn into very fine fibres that can be woven like any other yarn. Polyester is sold in the name of polycot, polywool, terrycot, etc. Polycot is a mixture of polyester and cotton; Polywool is a mixture of polyester and wool. PET (Polyethylene Terephthalate) is a very familiar form of polyester. It is used for making water and soda bottles, utensils, films and wires amongst many other useful products. Many of the clothes we wear are made out of polyester fibres. Fabrics made from this fibre do not get wrinkled easily and are easy to wash making polyester fabrics suitable for dress materials.

Fill the data in the following table

S.No	Type of thread/fibre	Total weight required to break the thread (in g)
1	Cotton	
2	Wool	
3	Silk	
4	Nylon	

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